

Job Market Intelligence Dashboard

This project transforms raw job-posting data into actionable insights for job seekers and employers. Using **Python** for data cleaning, analysis, and machine learning – and **Power BI** for interactive visualization – the dashboard covers three core areas: **Job Market Overview**, **Salary Intelligence**, and **Skills & Industry Intelligence**.

DATA ANALYSIS PYTHON MACHINE LEARNING POWER BI

Made with GAMMA

Project Idea: Turning Data Into Career Decisions

For Job Seekers

- Identify high-demand positions and required experience levels
- Compare salary differences across locations and work types
- Discover the most important skills for your target role
- Find where job opportunities are geographically concentrated

For Employers

- Understand hiring trends and seasonal patterns
- Benchmark work types and experience-level demand
- Identify which industries and companies are actively hiring
- Align job requirements with the skills currently in demand

By bridging raw data with interactive dashboards, this project empowers both sides of the labor market to make smarter, evidence-based decisions.

Made with GAMMA

What We Did in Python

01

Explore & Assess

Examined dataset structure, identified missing values, checked for duplicates, and evaluated overall data quality before any transformation.

02

Clean & Enrich

Removed inconsistencies and engineered new features covering job postings, salaries, experience levels, locations, work types, views, applications, and skills.

03

Analyze & Model

Performed statistical analysis on job market patterns and trained a machine learning model to estimate salaries based on job characteristics.

04

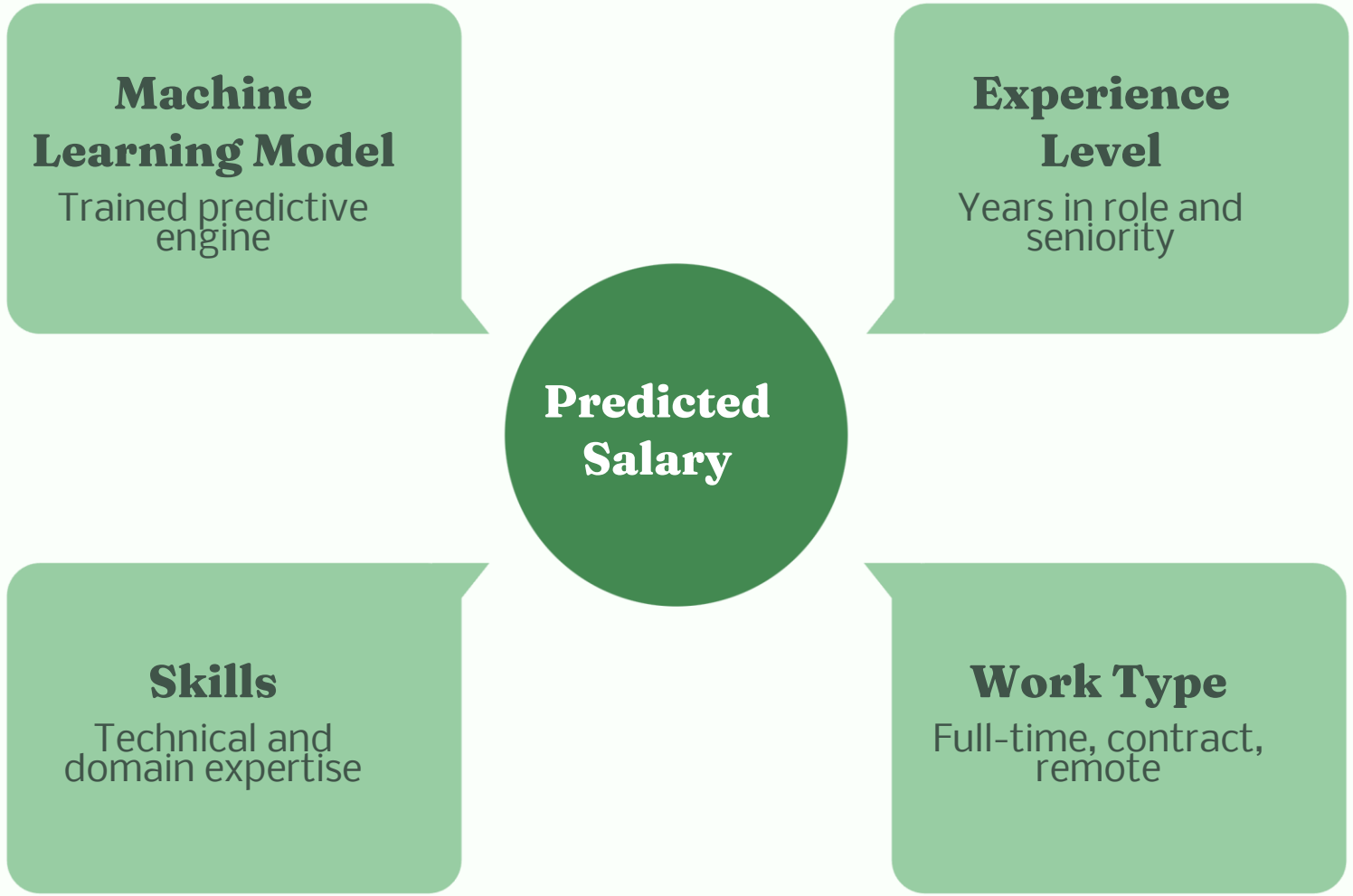
Visualize in Power BI

Exported processed data to Power BI and built three interactive dashboard pages covering market overview, salary intelligence, and skills insights.



Salary Prediction Model

Machine Learning for Career Intelligence



Made with GAMMA

Count of job id by day

? Question

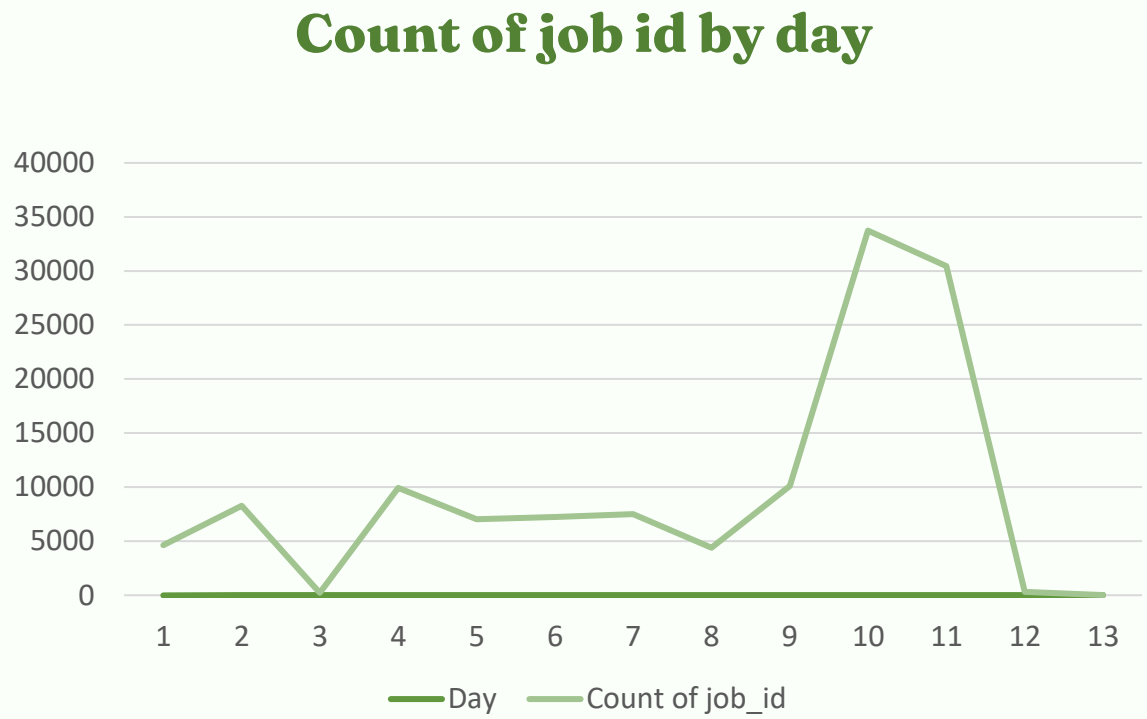
Which days have the highest and lowest job activity?

Insight

Job activity peaks around Day 18 and drops sharply after Day 19

Recommendation

Increase resources on high-demand days and investigate the sudden drop after Day 19.



Made with GAMMA

Count of the job by work type

? Question

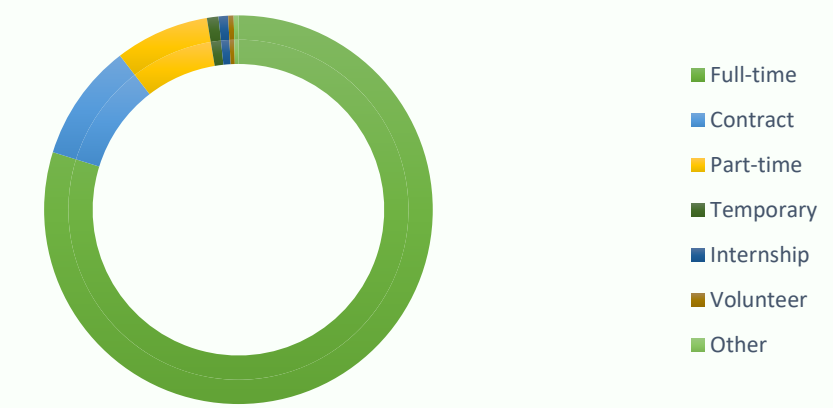
Which work type has the highest number of jobs?

Insight

Full-time jobs dominate at 79.78% of the total.

Recommendation

Focus on full-time recruitment while increasing opportunities for other work types.



Made with GAMMA

Jobs by title

? Question

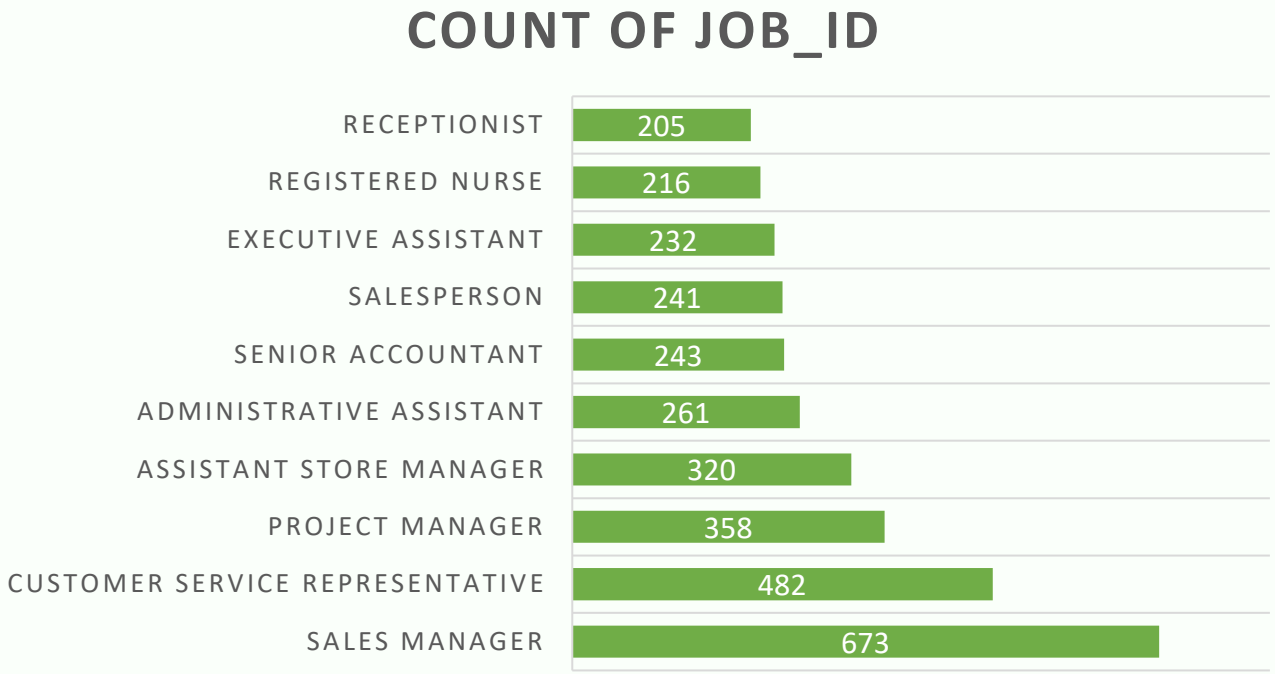
Which job titles have the highest number of job postings?

Insight

Sales Manager has the highest number of postings, followed by Customer Service Rep.

Recommendation

Prioritize recruitment for high-demand roles, especially Sales Manager positions.



Made with GAMMA

Job by experience level

? Question

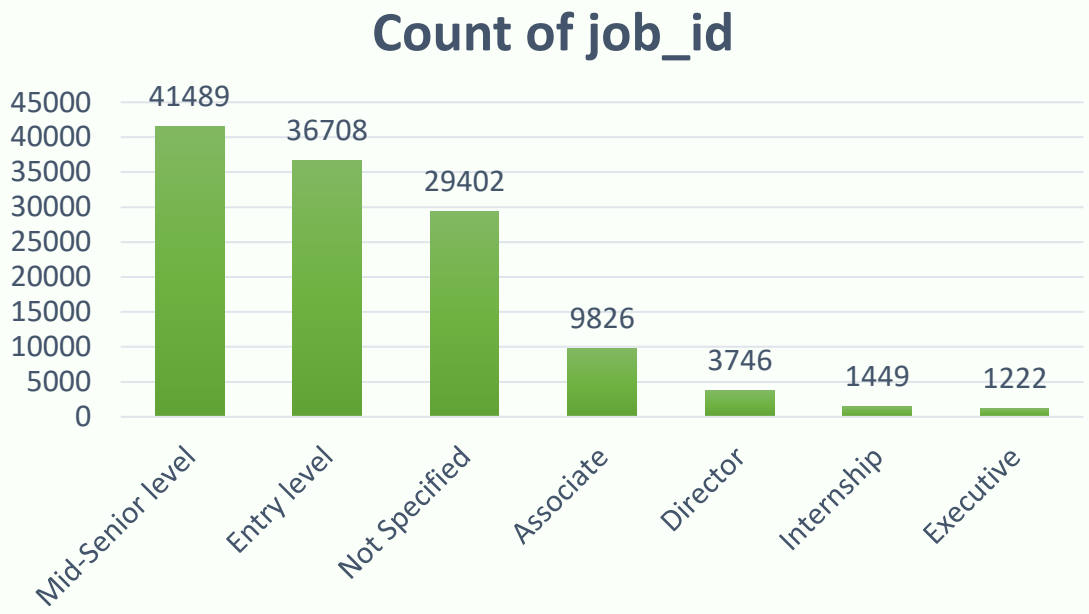
Which experience level has the highest number of job postings?

Insight

Mid-Senior Level has the highest demand, followed by Entry Level

Recommendation

Focus recruitment efforts on mid-senior and entry-level positions.



Made with GAMMA

Salary by Experience Level

? Question

How does salary differ by experience level?

Insight

Executive has the highest median salary, while Mid-Senior has the highest total average salary.

Recommendation

Focus compensation strategies on senior roles and create clear progression paths for junior employees.



Made with GAMMA

Job Market KPIs

? Question

What are the key overall statistics of the job market dataset?

Insight

The dataset contains 133.84K jobs across 24.69K companies and 8.53K locations.

Recommendation

Use these KPIs to monitor job volume, company participation, geographic coverage, and salary levels

123,842K

Count of job id

24,469K

Count of company id

8,526K

Count of location

22,02K

Average of med salary

Made with GAMMA

Salary by Remote Work

? Question

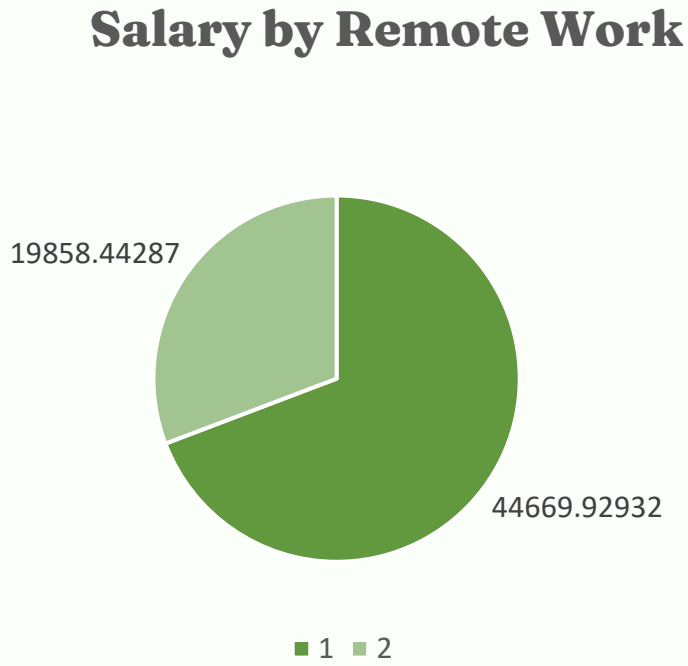
Does remote work affect average salary?

Insight

Remote jobs have a higher average salary (44.67K) than non-remote jobs (19.86K).

Recommendation

Expand remote opportunities to attract higher-paying roles and talent.



Made with GAMMA

Total Median Salary by Location

? Question

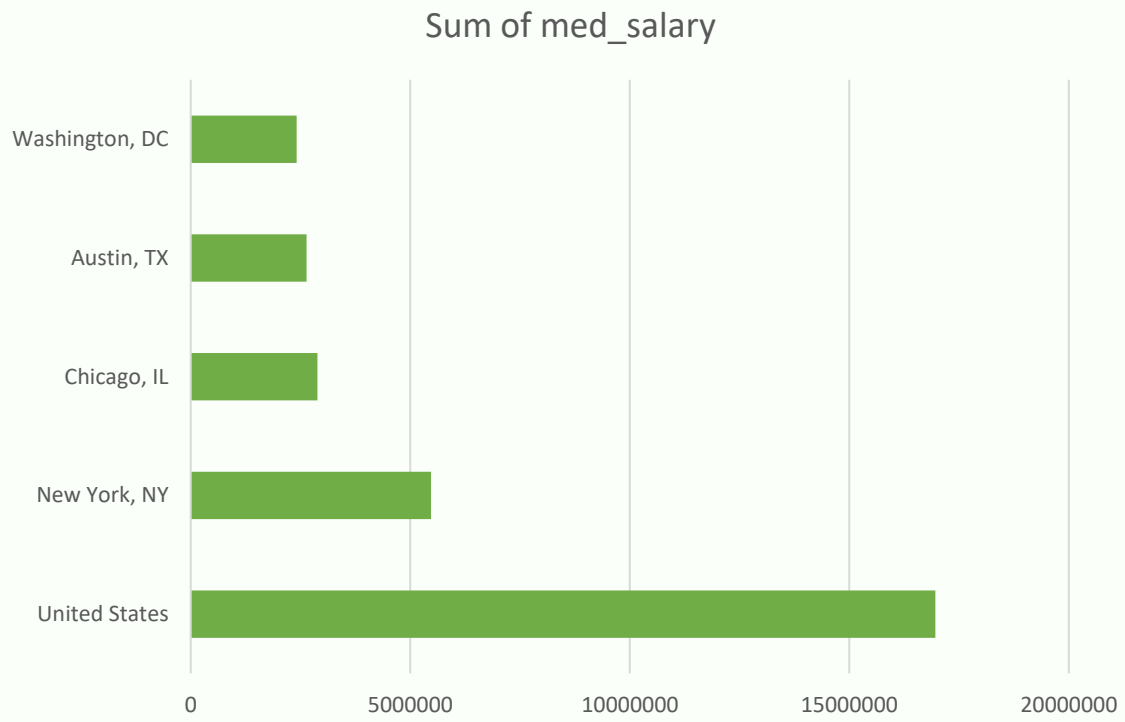
Which location has the highest total median salary?

Insight

The United States has the highest salary total, followed by New York

Recommendation

Focus recruitment and salary planning on the highest-salary locations



Made with GAMMA

Conclusion: From Raw Data to Real Insights

The analysis reveals a dynamic job market with clear structured trends: **Full-time roles dominate the market at 79.78%** of total postings, with a massive dataset comprising over **123.8K jobs** across **24.4K companies** and **8.5K locations**.

🔎 Demand Patterns

Mid-Senior and Entry-level roles drive the market demand, with **Sales Manager**, **Customer Service Rep**, and **Project Manager** emerging as the most posted positions.

💰 Salary & Location

Executive roles yield the highest median salary (~$200K), with major earning totals centered in top locations like **New York, NY ($5M+)**, Chicago, and Austin.

🧠 Skills & Work Style

Remote roles offer a significantly higher average salary (**$44.67K**) compared to non-remote options (**$19.86K**).

These insights help job seekers target high-value roles while empowering employers to refine recruitment and pay structures.

Made with GAMMA